

Multi-Agent Reinforcement Learning for Base Station Placement in TDOA-Based Localization

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Abstract—Accurate localization of devices is a key capability for emerging 5G and 6G networks and requires efficient placement strategies for base stations (BSs). Conventional geometry-based approaches such as Geometric Dilution of Precision (GDOP) ignore realistic propagation effects, in particular Non-Line of Sight (NLOS) conditions caused by buildings and other obstacles. This paper proposes a ray-tracing-assisted Multi-Agent Reinforcement Learning (MARL) framework for environment-aware BS placement in Time Difference of Arrival (TDOA) localization systems. Proximal Policy Optimization (PPO) agents are trained on Channel Impulse Responses (CIRs) generated from a detailed 3D model of a university campus. Each agent cooperatively places one BS while optimizing a shared reward that combines localization accuracy and coverage. The approach is evaluated on five campus segments with varying propagation characteristics. Results show that it achieves up to 11% reduction in localization Mean Absolute Error (MAE) compared with conventional GDOP-based placement, with the largest gains in segments with complex propagation conditions. These findings demonstrate that incorporating site-specific propagation data into the placement process yields substantial improvements over purely geometric strategies.

Index Terms—Base station placement, multi-agent reinforcement learning, ray tracing, sensor networks, TDOA localization

I. INTRODUCTION

Precise localization is a key feature for 5G/6G verticals such as Industry 4.0, asset tracking, and Non-Public Network (NPN) [1]. As sensing and communication converge into equal network functionalities in 6G architectures, network-based positioning becomes a core requirement [2], [3]. Time Difference of Arrival (TDOA) is an established method for network-based localization that requires no User Equipment (UE) synchronization and directly exploits existing base station (BS) infrastructure [4]. Among the design parameters that impact localization accuracy, BS placement plays a dominant role [5]. Geometric Dilution of Precision (GDOP), derived from the Fisher Information Matrix, is the standard metric for quantifying this geometric dependency [6]. However, GDOP captures purely geometric configuration and neglects propagation effects such as Non-Line of Sight (NLOS) propagation, multipath-induced Time of Arrival (TOA) bias, and path loss [7]. As a result, localization accuracy deviates systematically from GDOP-based predictions, and exhaustive

search for optimal BS placement remains infeasible due to combinatorial complexity $\mathcal{O}\left(\binom{M}{N}\right)$.

GDOP-based placement optimization is well established in Global Navigation Satellite System (GNSS) and radar networks [8], where predominantly Line of Sight (LOS) propagation permits purely geometric objective functions. These approaches do not transfer to cellular deployments, where NLOS components fundamentally change the effective measurement geometry. Heuristic methods such as Genetic Algorithm (GA) and Simulated Annealing (SA) have been applied to localization infrastructure planning [9] and measurement location planning for radio environment maps [10], but exhibit limited scalability and retain geometric objective functions, leaving the core problem unresolved. Deep Reinforcement Learning (DRL) has been explored for network planning targeting coverage and throughput [11] as well as for Unmanned Aerial Vehicle (UAV) BS positioning [12], yet neither formulation addresses localization accuracy as an optimization objective. The work in [13] investigates Reinforcement Learning (RL)-based BS placement with a TDOA-oriented objective, but relies on a simplified noise model and omits site-specific propagation effects. Ray-tracing-based Channel Impulse Response (CIR) data has been used to capture the structure of Multipath Components (MPCs) relevant for TDOA localization [14], without extending the framework to placement optimization. To the best of our knowledge, no prior work has combined RL with ray-tracing-based propagation modeling to optimize BS placement for TDOA localization accuracy.

This raises the question whether Multi-Agent Reinforcement Learning (MARL) can leverage propagation effects to improve BS placement for TDOA-based localization compared to purely geometrical approaches based on GDOP. Since field measurements at all candidate positions are infeasible, ray-tracing provides the only practical means to obtain complete propagation data for every BS–UE combination.

This work proposes a MARL framework in which agents cooperatively place BSs on a discrete 3D candidate grid. Policy training is conducted on CIR data generated by Sionna RT [15], allowing the learned policy to implicitly capture NLOS regions and multipath structure without requiring an

explicit propagation model at inference time. The framework is evaluated on a realistic outdoor campus scenario with building-induced NLOS conditions.

While prior approaches tackle geometry or propagation in isolation, no existing method combines both for TDOA-inspired BS placement. This work addresses that gap with the following contributions:

- A cooperative MARL formulation for TDOA BS placement, where N agents jointly optimize a shared reward combining localization accuracy and coverage.
- Environment-aware placement through ray-tracing-based CIR training, enabling the policy to account for NLOS and multipath effects.
- Empirical evaluation across campus segments with varying propagation complexity, demonstrating up to 11% Mean Absolute Error (MAE) reduction compared to GDOP baselines.

The remainder of this paper is structured as follows. Section II introduces the signal model, TDOA localization, and the problem formulation. Section III describes the proposed method, followed by the experimental setup in Section IV. Section V presents and discusses the results, and Section VI concludes the paper.

II. SYSTEM MODEL

This section introduces the signal model, derives the TDOA localization procedure, and formulates the BS placement optimization problem.

A. Signal Model

The wireless channel between a BS and a UE is characterized by its CIR, which describes the superposition of all propagation paths between the two locations. For BS i and a given UE, the CIR is modeled as a sum of L_i MPCs:

$$h_i(t) = \sum_{l=1}^{L_i} \omega_{i,l} \cdot \delta(t - \tau_{i,l}), \quad (1)$$

where $\omega_{i,l}$ and $\tau_{i,l}$ denote the complex amplitude and propagation delay of the l -th path of BS i , respectively. The TOA is estimated as the delay of the strongest received component:

$$\hat{\tau}_i = \tau_{i,l^*}, \quad l^* = \arg \max_l |\omega_{i,l}|. \quad (2)$$

Under LOS conditions this estimate coincides with the direct-path delay, but it incurs a positive bias whenever the direct path is obstructed. In this work, CIRs are obtained from deterministic ray-tracing, which models the site-specific MPCs including reflections, diffractions, and NLOS shadowing. The resulting NLOS-induced bias is thus captured in the data, unlike purely geometry-based metrics such as GDOP, which inherently ignore propagation effects.

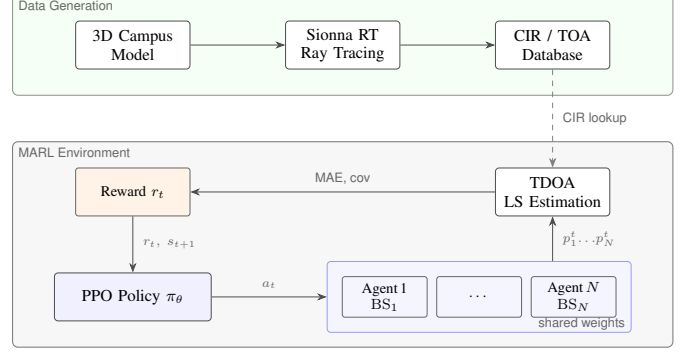


Fig. 1: System overview. Sionna RT generates a CIR database from the 3D model. N MARL agents cooperatively place BS based on a shared reward combining TDOA localization accuracy and coverage.

B. TDOA-based Localization

Given N BSs at known positions $\mathbf{b}_i \in \mathbb{R}^3$ and a UE at unknown position $\mathbf{p} \in \mathbb{R}^3$, the TDOA measured between BS i and a reference BS 1 is:

$$\Delta\tau_{i,1} = \frac{1}{c} (\|\mathbf{p} - \mathbf{b}_i\| - \|\mathbf{p} - \mathbf{b}_1\|) + n_{i,1}, \quad (3)$$

where c is the speed of light and $n_{i,1}$ denotes the measurement noise, comprising receiver thermal noise, synchronization errors between BSs, and NLOS-induced bias from obstructed propagation paths. Each observation constrains the UE to a hyperboloid, and jointly solving $N - 1$ such constraints yields the position estimate. For 2D localization with fixed UE height, at least $N \geq 3$ BSs are required.

C. Localization Error and Coverage

The 2D position estimate $\hat{\mathbf{p}} = (\hat{x}, \hat{y})$ is obtained via Least-Squares (LS) multilateration, and the localization error is the Euclidean distance to the true position:

$$e = \|\hat{\mathbf{p}} - \mathbf{p}\|_2. \quad (4)$$

Not every UE can be localized, since detectability requires valid TOA measurements from at least three BSs. Non-detectable UEs are excluded from the MAE computation, and their fraction is reported as coverage separately, capturing a distinct dimension of placement quality that MAE alone does not reflect.

D. Problem Formulation

Let $\mathcal{G} = \{g_1, \dots, g_M\}$ be the set of M discrete candidate BS positions on a 3D grid spanning the deployment area (the concrete grid dimensions and height levels are specified in Section IV). The task is to select a subset $\mathcal{B} \subset \mathcal{G}$ of N positions that jointly minimize localization error and non-coverage:

$$\mathcal{B}^* = \arg \min_{\mathcal{B} \subset \mathcal{G}} [\beta \cdot \text{MAE}(\mathcal{B}) + (1 - \beta) \cdot m(\mathcal{B})], \quad (5)$$

where

$$\text{MAE}(\mathcal{B}) = \frac{1}{|\mathcal{P}_d|} \sum_{\mathbf{p} \in \mathcal{P}_d} e(\mathbf{p}, \mathcal{B}) \quad (6)$$

is the mean localization error over the detectable UEs $\mathcal{P}_d \subseteq \mathcal{P}$, $m(\mathcal{B}) = (|\mathcal{P}| - |\mathcal{P}_d|)/|\mathcal{P}|$ is the non-coverage fraction, and $\beta \in [0, 1]$ controls the trade-off between the two objectives. With $\mathcal{O}\left(\binom{M}{N}\right)$ possible subsets, exhaustive search becomes computationally infeasible even for moderate M and N , motivating a learned optimization approach.

III. PROPOSED METHOD

We formulate the BS placement task as a sequential decision problem and solve it with cooperative MARL. This multi-agent formulation decomposes the joint placement of N BSs into N parallel decisions with shared learning. Each of the N agents controls one BS, and all agents share a common reward combining localization accuracy and coverage. The placement policy is optimized using Proximal Policy Optimization (PPO) with parameter sharing across agents. Fig. 1 presents the overall system architecture.

A. MDP Formulation

We formulate the BS placement problem as a cooperative multi-agent Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, R, \gamma)$, where γ denotes the discount factor (see Table 2), with N agents acting simultaneously to optimize the joint placement of N BSs.

1) *State space*: Let $\mathcal{G} \subset \mathbb{R}^3$ denote the set of placeable candidate positions. The state at time step t is the joint position of all N agents:

$$s_t = (p_1^t, \dots, p_N^t), \quad p_i^t \in \mathcal{G}. \quad (7)$$

Agents are initialized at randomly sampled positions $p_i^0 \in \mathcal{G}$ without replacement.

2) *Action space*: Each agent i maintains a current position $p_i^t \in \mathcal{G}$ at time step t and selects an action from its valid action set

$$\mathcal{A}_i^t = \{a \in \mathcal{N}(p_i^t) \mid a \notin \{p_j^t\}_{j \neq i}\}, \quad (8)$$

where $\mathcal{N}(p_i^t) \subseteq \mathcal{G}$ denotes a local neighborhood of p_i^t . Actions leading to positions occupied by other agents are excluded via action masking.

3) *Transition*: The transition function $\mathcal{T}$ is deterministic. All agents act simultaneously based on a shared occupancy snapshot taken at the beginning of each step. Each agent i moves to its selected target position $p'_i \in \mathcal{A}_i^t$. In case of conflicts, the agent with the lowest index has priority while the remaining agents stay at their current position. The subsequent state $s_{t+1} = (p_1^{t+1}, \dots, p_N^{t+1})$ reflects the updated agent locations.

4) *Reward*: All agents receive a shared team reward at each time step t :

$$r_t = 2(\beta \cdot r_{\text{loc}} + (1 - \beta) \cdot r_{\text{cov}}), \quad r_t \in [-2, 2], \quad (9)$$

where $\beta \in [0, 1]$ controls the trade-off between localization accuracy and coverage, consistent with the objective in Eq. (5). Both components use a tanh mapping:

$$r_{\text{loc}} = 1 - 2 \tanh\left(\frac{\text{MAE}(\mathcal{B})}{\sigma_{\text{loc}}}\right), \quad r_{\text{cov}} = 1 - 2 \tanh\left(\frac{m(\mathcal{B})}{\sigma_{\text{cov}}}\right), \quad (10)$$

where $\text{MAE}(\mathcal{B})$ is the mean localization error and $m(\mathcal{B})$ the fraction of UEs with insufficient TOA measurements for the current BS configuration $\mathcal{B}$. The scaling parameters σ_{loc} and σ_{cov} control the sensitivity of each component.

B. Network Architecture

The policy network processes the voxelized grid state s_t using a 3D Convolutional Neural Network (CNN) backbone, which extracts spatial features from the occupancy representation of the environment. For each agent, a region-of-interest of size $k=5$ voxels around its current position is extracted and mapped to a compact 64-dimensional token encoding both the local spatial context and the agent's position within the grid.

Each agent's token is passed to a shared actor head, which produces logits over all M candidate positions. Actions corresponding to occupied positions are suppressed via action masking before the softmax, ensuring that only valid placements are sampled. A centralized critic aggregates globally pooled features from all agents to produce a scalar value estimate, following the Centralized Training with Decentralized Execution (CTDE) paradigm. All agents share the same network weights, reducing the number of trainable parameters and improving sample efficiency.

The policy is optimized using PPO with a clipped surrogate objective and Generalized Advantage Estimation (GAE). Since all agents receive the same shared team reward, the joint log-probability of all agents' actions is used during backpropagation.

IV. EXPERIMENTAL SETUP

We first describe the simulation environment and its segmentation, then detail the RL training configuration and the baseline methods used for comparison.

A. Scenario and Ray-Tracing Simulation

Experiments are conducted on a model of the campus "Technische Fakultät" at FAU Erlangen-Nürnberg in Erlangen, Germany, shown in Fig. 2, covering approximately $790 \text{ m} \times 1090 \text{ m}$. Building geometries were extracted from OpenStreetMap¹ and imported into Sionna [15] via a Blender model. Multiple multi-story buildings produce NLOS shadowing, representative of a campus NPN deployment [1]. CIRs are generated at a carrier frequency of 3.75 GHz, corresponding to the German campus network band, for all combinations of $M \approx 300$ BS candidate and $N_{\text{UE}} \approx 350$ UE positions per segment. BS candidates are arranged on

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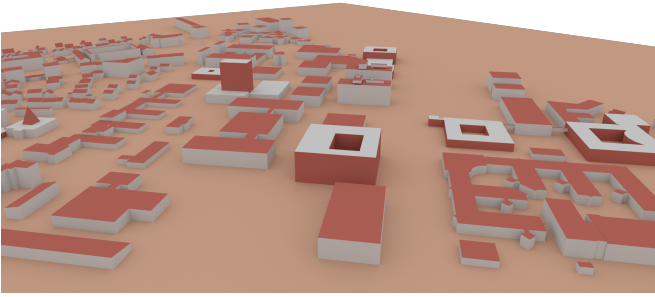


Fig. 2: 3D model of FAU Erlangen-Nürnberg Campus in Erlangen, Germany, created from OpenStreetMap data and imported into Sionna.

Tab. 1: Selected campus segments and their characteristics. Vis.: average fraction of BS candidates reachable from each UE position.

Segment	# UEs	# BS Cand.	Vis. (%)
S4	300	244	67.8
S6	359	332	86.5
S9	313	290	74.3
S11	386	343	88.1
S14	423	338	93.3

an equidistant grid with 20 m spacing at three height levels (1 m, 9 m, 17 m). UEs occupy a 10 m-spaced grid at a fixed height of 1.5 m. In both cases, positions inside buildings are excluded. A sampling frequency of 1 GHz provides a delay resolution of 1 ns, supporting robust TOA extraction from the 100 MHz bandwidth-limited CIR. TOA estimates are obtained from the strongest MPC, which introduces a deterministic positive bias under NLOS conditions [14]. UE positions are estimated via LS-based TDOA multilateration [4], requiring at least three TOA measurements per UE.

B. Environment Segmentation

The campus area is divided into $4 \times 4 = 16$ non-overlapping segments of approximately $198 \text{ m} \times 273 \text{ m}$ each. This size ensures that each segment contains enough BS candidates and UE positions for meaningful placement optimization, while also keeping the combinatorial complexity of exhaustive baseline evaluation $\mathcal{O}\left(\binom{M}{N}\right)$ computationally feasible. Each segment maintains its own BS candidate grid, UE grid, and precomputed CIR lookup table. Five randomly selected segments (S3, S7, S8, S12, S16) serve as the training set and five as the evaluation set, summarized in Table 1. The evaluation segments cover a wide range of propagation conditions, from heavily obstructed (S4, visibility 67.8%) to predominantly open environments (S14, visibility 93.3%). To capture this variation, Table 1 reports the visibility ratio, defined as the average fraction of BS candidates reachable from a given UE position. The training and evaluation sets do not overlap, allowing to test generalization to unseen propagation conditions.

Tab. 2: PPO hyperparameter configuration.

Hyperparameter	Value
Learning rate α	10^{-4}
Discount factor γ	0.99
GAE parameter λ	0.95
Clip ratio ϵ	0.20
Entropy coefficient c_{ent}	0.02
Value function coefficient c_{vf}	1.0
PPO epochs per update K	4
Mini-batch size	256
Rollout length T	64
Max. gradient norm	0.5
Total training updates	10,000
Reward weight β	0.5
Loc. scaling σ_{loc}	91
Cov. scaling σ_{cov}	0.5

C. Reinforcement Learning Configuration

Throughout this work, $N = 5$ BSs are placed per segment. To improve generalization, each training episode draws a segment randomly from the training set. The per-agent action space allows movements within the local neighborhood $\mathcal{N}(p_i^t)$. Invalid and occupied locations are excluded via action masking. A checkpoint is saved every 50 updates, and the model achieving the highest 100-episode moving average return is additionally saved as the best model per seed. The reward weight $\beta = 0.5$ balances localization accuracy and coverage equally. Each reward component is scaled to the range $[-1, 1]$ using a tanh mapping with scaling parameters σ_{loc} and σ_{cov} . Specifically, $\sigma_{\text{loc}} = 91 \text{ m}$ places the zero-crossing at approximately 50 m, and $\sigma_{\text{cov}} = 0.5$ places it at a non-coverage fraction of $m \approx 0.27$, corresponding to a coverage of approximately 73%. The PPO hyperparameter configuration is given in Table 2.

D. Baselines

As two heuristic baselines, we implement GDOP mean and median ranking, which select the N BS candidate locations with the lowest average and median GDOP across all UE locations, respectively, for each segment. These baselines represent the geometric standard for TDOA localization and have been shown to outperform random and coverage-based heuristics [6], [7], [16]. Both baselines use the same UE and BS candidate grids as the MARL agent. Note that metaheuristic approaches such as GA or SA optimize the same geometric GDOP objective and would thus converge to similar placements as the GDOP baselines used here.

E. Evaluation Metrics and Statistical Testing

The primary evaluation metric is the MAE of the 2D position estimates across all UE locations within a segment. Additionally, we report the coverage rate, i.e., the fraction of UE locations with at least three valid TOA measurements for TDOA localization. The MAE is computed only over positions where the LS solver converges to a finite estimate. To ensure statistical robustness, we compute 95% confidence

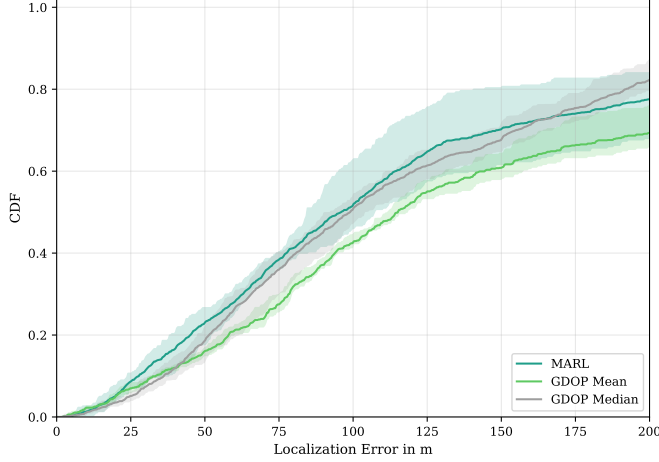


Fig. 3: Cumulative Distribution Function (CDF) of localization error for Segment S4, the most challenging evaluation segment with a visibility of 67.8%. The color-shaded areas represent the 95% confidence intervals.

intervals via bootstrap resampling with $B = 1000$ iterations over the random seeds.

V. RESULTS AND DISCUSSION

A. Results

Table 3 summarizes the localization performance across all five evaluation segments, none of which are used during training. On average, MARL achieves the lowest MAE of 89.6 m, compared to 93.3 m for GDOP Mean and 94.3 m for GDOP Median. MARL achieves the best MAE in three of five segments. The largest improvements occur in S9 and S4, where MARL reduces the MAE by approximately 11% for both segments (69.3 m vs. 78.0 m) and (79.9 m vs. 89.3 m). In S14, MARL achieves an 8% reduction (92.3 m vs. 100.0 m). In S6, all three methods perform comparably, while in S11 GDOP Median achieves the lowest MAE of 101.1 m. The CDF in Fig. 3 shows that MARL shifts the median error downward and reduces the upper tail of the distribution in S4, with the largest separation from both baselines between 25 and 125 m. The segment-wise error distributions in Fig. 4 confirm that MARL reduces the median error and interquartile range in S4 and S9. In S6, all three methods show largely overlapping distributions. In S11, MARL exhibits the widest spread, consistent with its higher MAE in this segment. Regarding coverage, GDOP Median achieves the highest average coverage (95%), followed by MARL (93%) and GDOP Mean (91%). In S9, the MAE improvement of MARL comes at the cost of reduced coverage (89% vs. 97%).

B. Discussion

The absolute MAE values of 69 m to 110 m reflect the challenging nature of the evaluation scenario. The limited number of $N = 5$ BSs per segment constrains the achievable geometric diversity. Additionally, since TOA estimates are derived from the strongest MPC, NLOS links introduce a

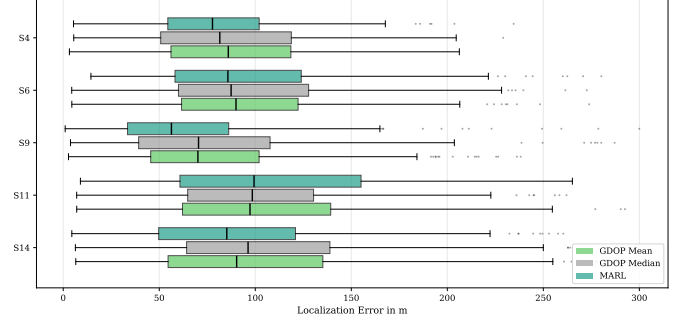


Fig. 4: Distribution of localization error across all evaluation segments.

systematic positive estimation bias that directly affects the TDOA position estimate. No NLOS mitigation is applied at the estimator level, as the focus of this work is on the relative improvement achievable through placement optimization rather than on minimizing absolute localization error.

While MARL achieves the lowest average MAE, the advantage varies across segments. Geometry-based GDOP treats all BS-UE links as equivalent and optimizes solely for geometric diversity. In contrast, the MARL policy is trained on ray-tracing-derived CIRs, enabling it to learn to avoid BS positions that appear geometrically favorable but predominantly produce NLOS-biased TOA estimates. This behavior is most evident in S4 and S9 (visibility 67.8% and 74.3%), where the divergence between geometric and actual propagation conditions is largest, and MARL achieves the strongest improvements. In S14 (visibility 93.3%), MARL still outperforms both baselines, suggesting that propagation-aware placement can also benefit predominantly open environments. In S11 (visibility 88.1%), however, GDOP Median achieves a lower MAE. One possible explanation is that S11 exhibits a more homogeneous propagation environment, where the geometric diversity captured by GDOP already closely approximates the actual measurement geometry, limiting the additional benefit of propagation-aware placement. This suggests that the advantage of MARL depends not on visibility alone but also on the spatial heterogeneity of the propagation conditions within each segment. The reduced coverage in S9 (89% vs. 97%) raises the question whether the MAE improvement is partly driven by selection bias, as non-detectable UEs are excluded from the error computation. However, the MAE advantage persists in S4 and S14, where coverage differences are small, indicating that the gain is not primarily a selection artifact. Instead, the coverage reduction in S9 suggests that the learned policy implicitly prioritizes placements with more reliable propagation paths over maximum detectability when both objectives are weighted equally.

Since all five evaluation segments are held out from training, the results indicate generalization to unseen propagation conditions. While the quantitative improvements are scenario-specific, validation across diverse environments is needed to confirm generalizability. Several limitations should be

Tab. 3: Localization performance across five campus segments ($N = 5$ BSs, 5 seeds). $\pm$ denotes the 95% bootstrap confidence interval ($B=1000$). Best MAE per segment is shown in **bold**.

Segment	GDOP Mean		GDOP Median		MARL (proposed)	
	MAE [m]	Cov. [%]	MAE [m]	Cov. [%]	MAE [m]	Cov. [%]
S4	89.3 $\pm$ 6.8	67	89.7 $\pm$ 6.2	85	79.9 $\pm$ 4.7	85
S6	96.3 $\pm$ 5.0	96	97.3 $\pm$ 5.7	95	96.9 $\pm$ 5.5	95
S9	78.0 $\pm$ 5.3	95	78.8 $\pm$ 5.9	97	69.3 $\pm$ 6.0	89
S11	103.0 $\pm$ 5.6	98	101.1 $\pm$ 4.9	98	109.6 $\pm$ 6.1	96
S14	100.0 $\pm$ 5.6	99	104.7 $\pm$ 5.2	98	92.3 $\pm$ 5.2	98
Avg.	93.3	91	94.3	95	89.6	93

noted. Hardware non-idealities such as receiver noise and synchronization errors may reduce the gains achievable in practice. Furthermore, the number of BSs is fixed at $N = 5$, and the sensitivity to this parameter is up for future work.

VI. CONCLUSION

In this paper, we proposed a MARL framework for optimizing BS placement in TDOA localization systems. By training on ray-tracing-generated CIRs from a 3D model of a university campus, the approach implicitly learns to account for effects of MPCs and NLOS conditions. Evaluation across five campus segments with varying propagation complexity shows that the MARL-based optimization achieves up to 11% reduction in localization MAE compared to conventional GDOP baseline methods, with the largest gains in S4 and S9, where visibility drops below 75%. However, the advantage is not uniform: in S11, the geometric baseline achieves lower error, and in S9 the MAE gain comes at the cost of reduced coverage (89% vs. 97%), suggesting that the accuracy–coverage trade-off warrants further investigation. Future work will focus on validating the proposed approach with field measurements, examining the impact of the number of BSs N on placement quality, and investigating generalization to environments with different propagation characteristics.

AUTHOR CONTRIBUTION

Conceptualization, B.P.; Methodology, B.P. and P.R.; Software, B.P. and P.R.; Validation, B.P.; Formal Analysis, B.P.; Investigation, B.P. and P.R.; Resources, B.P.; Data Curation, B.P. and P.R.; Writing – Original Draft Preparation, B.P.; Writing – Review and Editing, B.P., M.L. and N.F.; Visualization, B.P.; Supervision, N.F. and M.L.; Project Administration, N.F. and M.L.; All authors have read and agreed to the published version of the manuscript.

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